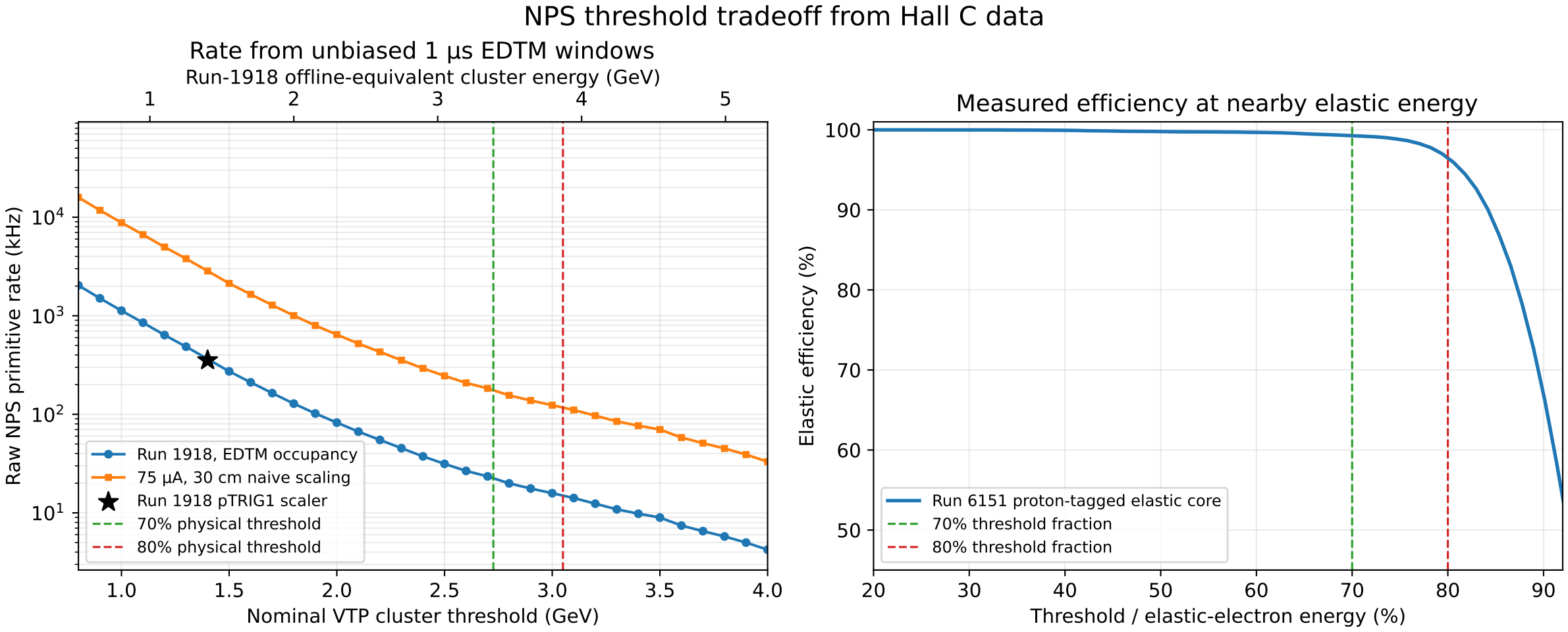
NPS threshold tradeoff from Hall C data
Rate from unbiased 1 µs EDTM windows
Run-1918 offline-equivalent cluster energy (GeV)
1
2
3
4
5
Raw NPS primitive rate (kHz)
10^4
10^3
10^2
10^1
Run 1918, EDTM occupancy
75 µA, 30 cm naive scaling
Run 1918 pTRIG1 scaler
70% physical threshold
80% physical threshold
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Nominal VTP cluster threshold (GeV)
Measured efficiency at nearby elastic energy
Elastic efficiency (%)
100
90
80
70
60
50
Run 6151 proton-tagged elastic core
70% threshold fraction
80% threshold fraction
20
30
40
50
60
70
80
90
Threshold / elastic-electron energy (%)